

NETRAJYOTI AI

High-Level Design (HLD) | Current Capstone Implementation and Future Controlled Scope

Scope statement

The current capstone is a Bengali-first deterministic eye-care guidance MVP. PostgreSQL supplies both routing criteria and RAG guidance. Java fixes the route. Local Ollama is optional, feature-flagged, and can only explain the already-fixed route after Java validation. Agent, MCP, skills registry, managed service directory, coordinator workflow, and persistent audit store remain future scope.

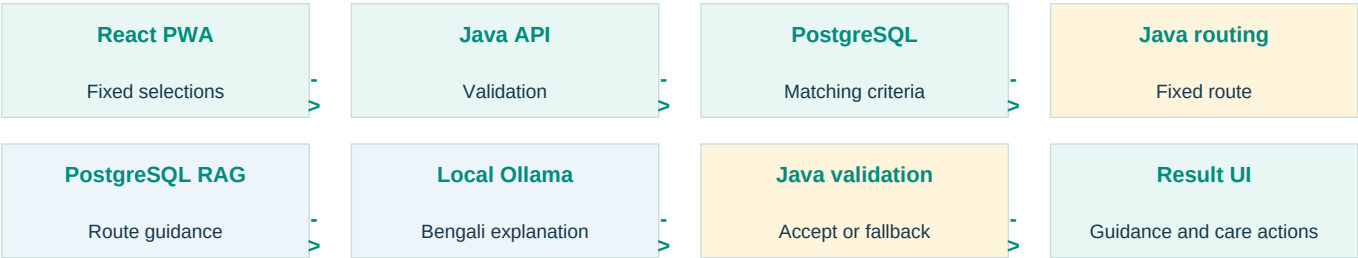
1. Architecture principles

Principle	Current implementation
Safety before automation	Java applies PostgreSQL criteria with fixed precedence: URGENT -> HUMAN_SUPPORT -> ROUTINE. Ollama never selects or changes a route.
Bengali-first access	React + TypeScript PWA provides bilingual controls, Bengali-first route guidance, optional browser voice input, typed fallback, and family-message support.
Privacy by design	Voice, free text, location, patient details, photos, and records do not influence routing or reach the Ollama endpoint. Patient details remain in browser memory until the user chooses to share.
Resilience	If PostgreSQL guidance retrieval, Ollama, validation, location, speech, or sharing is unavailable, the safe route and Java fallback guidance remain available.

2. Current MVP logical architecture

Layer	Component	Current responsibility
Experience	React + TypeScript PWA	Guided screening, local workflow state, selected symptoms, service direction, share/copy, and browser capabilities.
Application	Java 21 + Spring Boot REST API	Input validation, CORS/security configuration, deterministic routing, feature flags, Ollama orchestration, output validation, and safe fallback.
Safety domain	SafetyRoutingService	Retrieves matching PostgreSQL criteria and applies a fixed, reviewable route precedence.
Data	PostgreSQL + Spring JDBC/JPA	Stores clinical pathways, criteria, Bengali guidance, and source references. Current seeded records are DEMO_SIMULATED_NOT_FOR_CLINICAL_USE.
AI explanation	Local Ollama (optional)	Creates Bengali explanation from route-specific PostgreSQL guidance only after Java has fixed the route.
Care navigation	Frontend demo service options	Shows demo PHC, eye-care, ASHA, and family-support direction. It is not yet a PostgreSQL-managed verified directory.

Current capstone request flow



3. Deterministic routing and PostgreSQL criteria

The browser sends fixed concern codes and an explicit request for human support. Java asks PostgreSQL for eligible criteria that match those codes. In the capstone demo, optional Step 3A history is guidance context and does not override a route.

Condition	Java outcome
Any URGENT_IF_ANY criterion	URGENT
Otherwise, ESCALATE_IF_ANY criterion or explicit support request	HUMAN_SUPPORT
Otherwise, ROUTINE_IF_ANY criterion	ROUTINE
No eligible criteria	Conservative baseline or Human Support in explicit demo mode

Demo data status
Seeded pathways, criteria, and Bengali guidance are labelled DEMO_SIMULATED_NOT_FOR_CLINICAL_USE. They are available only when DEMO_RAG_ENABLED=true. They support the capstone demonstration but are not a clinical protocol or production-ready content release.

4. PostgreSQL RAG and local Ollama explanation

Stage	Actual current behaviour	Safety boundary
Feature flags	AI_SUMMARY_ENABLED enables Ollama; DEMO_RAG_ENABLED allows simulated demo records.	Both flags are backend-only configuration.
RAG retrieval	After route selection, PostgreSQL retrieves Bengali guidance and source data matching the fixed route.	Retrieved content grounds the explanation; it does not re-evaluate urgency.
Generation	Local Ollama (default llama3.2:3b) produces short Bengali guidance.	No raw voice, free text, patient details, precise location, photos, or records are sent.
Validation	CaregiverSummaryPolicy validates the result; the API records whether the output is validated or fallback.	Rejected, unavailable, or unsafe output becomes Java fixed Bengali guidance.
Presentation	The UI shows one source at a time: validated Ollama/RAG explanation or Java fallback.	Ollama cannot diagnose, prescribe, invent a facility, or alter the selected route.

5. Current data, service-navigation, and audit boundaries

- Clinical criteria, route guidance, and reference sources are in PostgreSQL. The database is the retrieval source for RAG guidance as well as the criteria repository for Java routing.
- Demo facility and support options are currently frontend demonstration data. A managed PostgreSQL service directory, freshness workflow, and coordinator administration interface are not implemented.
- The backend logs routing inputs, matched criteria, selected outcome, Ollama generation, validation, and fallback reasons. A persistent non-identifying audit-event store is future scope.
- Browser location, sharing, and speech are optional progressive capabilities; their absence must not block safety guidance or a human-support route.

6. Future controlled-pilot scope (not implemented in the current MVP)

Future capability	Future role	Required gate
Managed service directory and coordinator workflow	Maintain partner-verified facilities, ownership, availability, and freshness metadata.	Partner data source, named owner, human approval, and rollback.

Future capability	Future role	Required gate
Persistent audit and evaluation store	Capture non-identifying correlation, version, latency, validation, and fallback information.	Privacy design, retention policy, and operational monitoring.
Agent, Skills, and MCP gateway	Bounded, server-side retrieval or operations support downstream of the fixed route.	Allowlisted tools, least privilege, review, testing, and no autonomous action.
Alternative model providers	A future provider choice may replace or supplement local Ollama.	Clinical/content review, privacy approval, Bengali evaluation, and failure tests.

7. Deployment view

Zone	Current capstone implementation	Future controlled scope
Frontend	React PWA runs locally for the demo; GitHub Pages publishes the wireframe documentation.	Production HTTPS hosting with deployment monitoring.
Backend	Spring Boot API runs locally outside Docker during the demo.	Managed deployment, secret manager, operational monitoring, and scaling.
Data and AI	Docker Compose runs PostgreSQL and local Ollama. Ollama is not exposed directly to the browser.	Managed PostgreSQL, approved content lifecycle, optional model-provider gateway, and audited controls.
External capabilities	Browser speech, share, and location are optional and have safe alternatives.	Partner-verified read-only service data integrations only after governance approval.

8. Capstone acceptance statement

HLD decision

The current capstone uses PostgreSQL-backed deterministic routing and PostgreSQL RAG retrieval. Java fixes the route before any model call. Local Ollama improves Bengali communication only after the route is fixed and is subject to Java validation and fallback. The MVP does not diagnose disease, prescribe treatment, book care, contact users autonomously, or publish service changes.